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DTSC 2302

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Portuguese Language Student Performance Part 1

Six hundred students were observed. They were asked about their school, sex, age, address, family size, parent's cohabitation status, mother’s education, father’s education, mother’s job, father’s job, guardian, travel time, weekly study time, the number of past class failures, the quality of family relationships, free time after school, workday alcohol consumption, weekend alcohol consumption, current health status, the number of school absences, first period grade, second grade, and final grade . They had to answer whether they had extra educational support, family educational support, extra paid classes within the course subject, extra-curricular activities, attended nursery school, want to take higher education, have internet access at home, have a romantic relationship, hang out with friends, and why they choose their school. A prediction model using all these predictors was created.

The prediction model we made uses a multiple linear regression model that predicts students final grades (G3) using all 32 variables, including demographic factors, academic indicators, and family background. Looking closely at our results, the strongest predictor in the model is G2. A one point increase in second period grade has a relationship with an increase of around 0.864 points in the final grade, this shows strong persistence in academic performance. In comparison, for G1, a one point increase in first period grade increases G3 by about 0.138 points. Its effect is much smaller than G2, suggesting that recent performances are more influential.

Lastly, classes failed by the student reduce the expected final grade by about 0.125 points, meaning prior failures negatively impact future performance.

The overall quality of the model is strong, it explains a large part of variation in students' final grades. With an R^2 of 0.858 and an adj. R^2 of 0.850, the model accounts for about 85% of the variability in G3, indicating a good fit. The RMSE is about 1.22, suggesting that the model's predictions are close to the actual values. Additionally, the overall F-test is significant, meaning the model has useful predictive power. However, there are some limitations, many of the predictors are not statistically significant, which suggests that not all of them are contributing to predicting the final grades and the model could be more efficient, and simplified. As for out-of-sample performance, the model was only evaluated using the same data it was trained on. This means that the performance metrics may be optimistic, and the model may not perform as well on new data. Without using cross-validation or other techniques, it's hard to assess how well the model would generalize. So, while the model is useful for understanding relationships and making predictions within this sample, its reliability in the future is uncertain.

Appendix 1

OLS Regression Results

Dep. Variable:	G3	R-squared:	0.858
Model:	OLS	Adj. R-squared:	0.850
Method:	Least Squares	F-statistic:	107.0
Date:	Tue, 17 Mar 2026	Prob (F-statistic):	3.99e-217
Time:	16:00:32	Log-Likelihood:	-970.12
No. Observations:	600	AIC:	2006.
Df Residuals:	567	BIC:	2151.
Df Model:	32		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.0735	0.972	-0.076	0.940	-1.983	1.836
school	-0.2532	0.129	-1.960	0.051	-0.507	0.001
sex	-0.0891	0.123	-0.725	0.468	-0.330	0.152
age	0.0531	0.050	1.066	0.287	-0.045	0.151
address	0.1017	0.126	0.807	0.420	-0.146	0.349
famsize	0.0291	0.121	0.241	0.809	-0.208	0.266
Pstatus	-0.1525	0.176	-0.865	0.387	-0.499	0.194
Medu	-0.0405	0.068	-0.594	0.553	-0.174	0.093
Fedu	0.0326	0.064	0.508	0.611	-0.093	0.158
Mjob	0.0126	0.049	0.257	0.797	-0.083	0.108
Fjob	-0.1146	0.064	-1.796	0.073	-0.240	0.011
reason	-0.0888	0.045	-1.959	0.051	-0.178	0.000
guardian	-0.0178	0.109	-0.163	0.871	-0.232	0.196
traveltime	0.1354	0.076	1.790	0.074	-0.013	0.284
studytime	0.0459	0.069	0.669	0.504	-0.089	0.181
failures	-0.2152	0.100	-2.145	0.032	-0.412	-0.018
schoolsup	-0.2295	0.176	-1.302	0.194	-0.576	0.117
famsup	0.1459	0.111	1.315	0.189	-0.072	0.364
paid	-0.0619	0.231	-0.268	0.789	-0.516	0.392
activities	0.0461	0.108	0.427	0.670	-0.166	0.258
nursery	-0.0750	0.131	-0.573	0.567	-0.332	0.182
higher	0.3067	0.189	1.626	0.105	-0.064	0.677
internet	0.0585	0.133	0.440	0.660	-0.202	0.319
romantic	-0.0792	0.112	-0.709	0.478	-0.298	0.140
famrel	-0.0114	0.056	-0.204	0.838	-0.121	0.098
freetime	-0.0407	0.056	-0.729	0.466	-0.150	0.069
goout	-0.0273	0.052	-0.528	0.598	-0.129	0.074
Dalc	-0.0254	0.072	-0.352	0.725	-0.168	0.117
Walc	-0.0188	0.057	-0.331	0.741	-0.130	0.093
health	-0.0537	0.038	-1.425	0.155	-0.128	0.020
absences	0.0123	0.013	0.973	0.331	-0.012	0.037
G1	0.1378	0.039	3.545	0.000	0.061	0.214
G2	0.8641	0.036	24.064	0.000	0.794	0.935

Omnibus:	415.490	Durbin-Watson:	2.106
Prob(Omnibus):	0.000	Jarque-Bera (JB):	8881.538
Skew:	-2.749	Prob(JB):	0.00
Kurtosis:	21.029	Cond. No.	487.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.